CHARACTERIZATIONS OF MULTICROSS SINGULARITIES BY THE DELTA-INVARIANT AND BY ITERATED GLUEINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the imported algebraic equivalence, the completed-ring inclusion with the branches/delta additivity computation, the normalization factorization, and the squishing exclusion by contradiction are all retained verbatim.

Editorial note. The label lemma-multicross names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Multicross versus delta-invariant. Same chapter, Lemma lemma-multicross-algebra. Supplies the equivalence between the defining completed normal form and the delta-invariant equal to branches minus one.
- Glueing of two points and the almost-isomorphism factorization. Same chapter, Example example-glue-points and Lemma lemma-factor-almost-isomorphism. Defines the glueing operation and factors the normalization map into glueings and tangent-vector squishings.

Source reference. Original reference 0C1J: sources/stacks-curves/pinned-original-source.tex, source lines 3202–3202. Prerequisite pointer invoked at proof line(s) 3489; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1L: sources/stacks-curves/pinned-original-source.tex, source lines 3257–3257. Prerequisite pointer invoked at proof line(s) 3515; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1V: sources/stacks-curves/pinned-original-source.tex, source lines 3431–3431. Prerequisite pointer invoked at proof line(s) 3494; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be an algebraically closed field. Let X be a reduced algebraic 1-dimensional k-scheme. Let $x \in X$. The following are equivalent*

(1) *x defines a multicross singularity,*

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(2) *the δ-invariant of X at x is the number of branches of X at x minus 1,*
(3) *there is a sequence of morphisms $U_n \to U_{n-1} \to \ldots \to U_0 = U \subset X$ where U is an open neighbourhood of x, where U_n is nonsingular, and where each $U_i \to U_{i-1}$ is the glueing of two points as in Example 0C1J.*

Proof. The equivalence of (1) and (2) is Lemma 0C1V.

Assume (3). We will argue by descending induction on i that all singularities of U_i are multicross. This is true for U_n as U_n has no singular points. If U_i is gotten from U_{i+1} by glueing $a, b \in U_{i+1}$ to a point $c \in U_i$, then we see that

$$\mathcal{O}^\wedge_{U_i,c} \subset \mathcal{O}^\wedge_{U_{i+1},a} \times \mathcal{O}^\wedge_{U_{i+1},b}$$

is the set of elements having the same residue classes in k. Thus the number of branches at c is the sum of the number of branches at a and b, and the δ-invariant at c is the sum of the δ-invariants at a and b plus 1 (because the displayed inclusion has codimension 1). This proves that (2) holds as desired.

Assume the equivalent conditions (1) and (2). We may choose an open $U \subset X$ such that x is the only singular point of U. Then we apply Lemma 0C1L to the normalization morphism

$$U^\nu = U_n \to U_{n-1} \to \ldots \to U_1 \to U_0 = U$$

All we have to do is show that in none of the steps we are squishing a tangent vector. Suppose $U_{i+1} \to U_i$ is the smallest i such that this is the squishing of a tangent vector θ at $u' \in U_{i+1}$ lying over $u \in U_i$. Arguing as above, we see that u_i is a multicross singularity (because the maps $U_i \to \ldots \to U_0$ are glueing of pairs of points). But now the number of branches at u' and u is the same and the δ-invariant of U_i at u is 1 bigger than the δ-invariant of U_{i+1} at u'. By Lemma 0C1V this implies that u cannot be a multicross singularity which is a contradiction. $\square$